

Electrical Instrumentation Part-II

Two-Day Reverse Engineering Roadmap for Semester Examination

Built from the solved last-four-year Part-II pattern and uploaded module materials

Purpose of this roadmap: You do not need to become a complete expert in two days. You need to become a **semester-answer engineer**: able to recognise the question family, draw the correct labelled diagram, write the governing formula, derive the result block-by-block, and finish with a boxed answer in the examiner's expected style.

Diagram command for every answer: use straight and perpendicular lines wherever possible; keep labels far away from wires and components; never place a word on a line; if a diagram becomes crowded, split it into two smaller diagrams. A clean partial diagram is better than a crowded full diagram.

Contents

1	The Big Reverse Engineering Idea	4
2	Two-Day War Plan	6
2.1	Day 1: Build the instruments and feedback systems	6
2.2	Day 2: Build the conversion and filter-design engine	6
3	Examiner-Sense Answer Skeleton	8
3.1	For 2-mark questions	8
3.2	For 4-mark questions	8
3.3	For 6 to 8-mark questions	8
3.4	For 10-mark design questions	8
4	Diagram Drawing Command Sheet	10
4.1	Universal diagram rules	10
4.2	Safe TikZ block style	10
4.3	Safe signal-flow diagram template	10
5	Phase 1: CRO and Display-System Block	11
5.1	What to understand first	11
5.2	CRO question family	11
5.3	Grassroot derivation pattern for Lissajous phase	11

5.4	Clean CRO answer skeleton	12
6	Phase 2: Switched-Capacitor Circuits	13
6.1	What to understand first	13
6.2	What the examiner wants	13
6.3	SC question attack	13
7	Phase 3: State-Variable and Universal Filters	15
7.1	One-line concept	15
7.2	Standard denominator	15
7.3	State-variable output idea	15
7.4	Exam derivation pattern	15
8	Phase 4: PLL	17
8.1	One-line concept	17
8.2	Blocks of PLL	17
8.3	Derivation you must memorise	17
8.4	PLL demodulator logic	18
9	Phase 5: DAC and ADC	19
9.1	DAC minimum understanding	19
9.2	DAC error answers	19
9.3	ADC minimum understanding	19
9.4	ADC types	20
10	Phase 6: Butterworth and Chebyshev Filters	21
10.1	Butterworth: maximally flat	21
10.2	Chebyshev: equiripple passband	21
10.3	Why Chebyshev -3 dB frequency is greater than 1 rad/s	21
10.4	Band-pass and band-stop design	21
11	Question Recognition Table	23
12	Last-48-Hour Practice Protocol	24
12.1	Three-pass method	24
12.2	How to self-check	25
13	One-Page Formula Bank	26

14 Final Exam Writing Commands**27**

1. The Big Reverse Engineering Idea

In the semester paper, the examiner is not asking random theory. Almost every question comes from one of the following repeated machines:

Machine	What it does	Typical exam command
CRO and display system	Converts voltage signal into visible trace, measures time, phase and frequency	explain, draw, measure, justify
Switched-capacitor circuits	Replaces resistor by clocked capacitor; realizes filters in IC form	derive, replace, calculate, realize
State-variable filters	Gives LP, HP, BP, BS outputs from same active circuit	universal filter, transfer function
PLL	Locks VCO phase/frequency to input; demodulates or synthesizes frequency	stages, linear model, demodulator
DAC/ADC	Converts digital-to-analog and analog-to-digital	numericals, errors, flowchart
Filter approximation and design	Butterworth/Chebyshev poles, order, transfer function, realization	prove, compare, design

The reverse engineering method

When you see any question, do not start writing immediately. First identify the **question type**. Then use the matching skeleton.

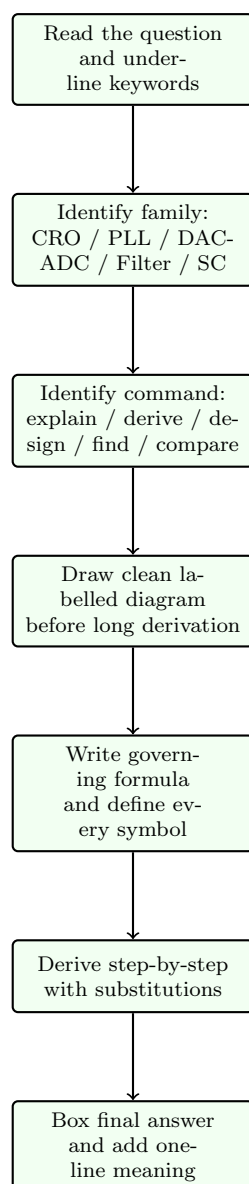


Figure 1: Reverse engineering algorithm for every Electrical Instrumentation Part-II question.

Your answer formula: Definition $\rightarrow$ Need/working principle $\rightarrow$ labelled diagram $\rightarrow$ equation $\rightarrow$ derivation $\rightarrow$ interpretation $\rightarrow$ boxed final result.

2. Two-Day War Plan

2.1 Day 1: Build the instruments and feedback systems

Time	Task	Output you must produce
6:30–7:00	Warm-up formula scan	Read only the formula bank in Section 12. No deep study. Just see the shapes.
7:00–9:00	CRO block	Draw CRO functional block, time-base, trigger, delay line, Lissajous ellipse, matched probe. Write one answer each in short form.
9:15–11:15	Switched capacitor	Master $R_{eq} = 1/(f_s C_s)$, resistor replacement, high-pass/band-pass idea, IC advantages. Redraw SC blocks cleanly.
11:30–1:00	State-variable filters	Learn why universal, outputs HP/BP/LP/BS, derive BP/BS denominator from two integrators.
2:30–4:30	PLL	Draw PLL block, linear model, lock/capture range, frequency demodulator, synthesizer formula.
5:00–7:00	Write one full 8-mark answer	Choose PLL linear model or state-variable filter. Write from zero with diagram and derivation.
8:30–10:30	Mixed PYQ attack	Solve one CRO, one SC, one PLL question without seeing notes. Then compare with solved files.

2.2 Day 2: Build the conversion and filter-design engine

Time	Task	Output you must produce
6:30–8:30	DAC	R-2R ladder, weighted sum formula, LSB, linearity error, offset error, settling time, monotonicity.
8:45–10:45	ADC	SAR flowchart, counter ADC, dual-slope ADC, sample-and-hold, quantization error, conversion time.
11:00–1:00	Butterworth and Chebyshev	Unit circle, ellipse, ripple, -3 dB issue, order formula, pole meaning.
2:30–5:00	Filter design numericals	Low-pass to band-pass, band-stop from band-pass, $Q = \omega_0/B$, passband gain.

5:30–7:30	Full paper simulation	Pick one last-year Part-II paper. Write skeleton answers for all questions.
8:30–10:30	Final compression	Make one page of diagrams and one page of formulas. Sleep after revising boxed results.

Strict rule for the last night: Do not try to read new theory after 10:30 pm. Revise diagrams and boxed formulas. In the exam, a clean labelled diagram plus correct governing formula often gives the first half of the marks.

3. Examiner-Sense Answer Skeleton

3.1 For 2-mark questions

Use four lines only:

1. Definition in one sentence.
2. Formula or key condition.
3. One physical meaning.
4. Boxed final result if numerical.

3.2 For 4-mark questions

1. Definition.
2. Small labelled diagram.
3. Formula with symbols.
4. Short derivation or explanation.
5. Final conclusion.

3.3 For 6 to 8-mark questions

1. Start with direct answer: “The circuit is used because...” or “The filter is called universal because...”
2. Draw diagram before derivation.
3. Write all assumptions.
4. Derive in small algebraic steps.
5. Compare with standard form.
6. Box the important parameters.
7. Add physical interpretation.

3.4 For 10-mark design questions

Step 1: Given data and unit conversion.

Step 2: Select standard transfer function.

Step 3: Calculate order or Q .

Step 4: Substitute numerical values.

Step 5: Choose realization topology.

Step 6: Draw circuit/block diagram.

Step 7: Verify gain, cut-off, bandwidth or centre frequency.

Your exam voice: write like a teacher is checking quickly. Put headings. Keep equations centred. Put the final numerical value in a box. Do not hide the answer inside a paragraph.

4. Diagram Drawing Command Sheet

4.1 Universal diagram rules

Rule 1: Use horizontal signal flow from left to right.

Rule 2: Use vertical branches only for feedback, ground, or control path.

Rule 3: Put labels above or below wires, never on the wire.

Rule 4: For long labels, split into two or three lines: recovered displacement, phase-sensitive demodulator.

Rule 5: If the diagram has more than six blocks, split it into two figures.

Rule 6: For circuits, connect only to wire junctions, not on capacitor plates or resistor bodies.

Rule 7: Put dot at every junction where branches meet.

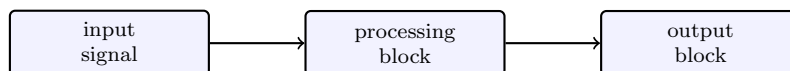
Rule 8: Keep captions short and meaningful.

4.2 Safe TikZ block style

Use this style in your own LaTeX diagrams:

```
block/.style={draw, rectangle, thick, minimum width=2.5cm,  
minimum height=0.85cm, text width=2.25cm, align=center,  
font=\scriptsize, inner sep=4pt}
```

4.3 Safe signal-flow diagram template



Label placed away from wire

Figure 2: Safe block diagram style with separated labels.

5. Phase 1: CRO and Display-System Block

5.1 What to understand first

A CRO converts voltage into deflection. The vertical system represents signal amplitude, while the horizontal system represents time. Therefore, every CRO answer is built from this sentence:

CRO core idea: vertical deflection $\propto$ input voltage, horizontal deflection $\propto$ time-base voltage. A stable trace needs triggering, and correct high-frequency measurement needs a matched probe.

5.2 CRO question family

Question word	key-	What to draw	Formula or conclusion
Functional blocks		vertical amplifier, delay line, trigger, time-base, horizontal amplifier, CRT	each block function in 2 lines
Trigger input		trigger circuit feeding time-base generator	starts sweep at same level and same slope
Delay line		vertical path delayed before plates	shows leading edge of waveform
Time-base generator		capacitor charging and fast discharge	$V_x(t) = kt$ during trace
Lissajous phase		ellipse in X-Y mode	$\phi = \sin^{-1}(y_1/y_2)$
Probe compensation		probe $R_p \parallel C_p$ and CRO $R_s \parallel C_s$	$R_p C_p = R_s C_s$
Frequency by Lissajous		stationary pattern tangencies	$f_y/f_x = N_h/N_v$

5.3 Grassroot derivation pattern for Lissajous phase

$$x = x_0 \sin \omega t, \quad y = y_0 \sin(\omega t + \phi)$$

At Y-axis intercept, $x = 0$:

$$\sin \omega t = 0 \Rightarrow \cos \omega t = \pm 1$$

Then

$$y = \pm y_0 \sin \phi$$

Hence

$$\boxed{\sin \phi = \frac{y_1}{y_2}}, \quad \boxed{\phi = \sin^{-1} \left(\frac{y_1}{y_2} \right)}$$

5.4 Clean CRO answer skeleton

Block 1: Define CRO in one sentence.

Block 2: Draw the required diagram.

Block 3: Explain signal path: input $\rightarrow$ amplifier $\rightarrow$ plates $\rightarrow$ screen.

Block 4: Write formula.

Block 5: Explain what happens if the block is absent.

Block 6: Box final measurement relation.

6. Phase 2: Switched-Capacitor Circuits

6.1 What to understand first

A switched capacitor is a resistor made out of charge transfer. During one clock phase the capacitor samples charge; during the next phase it delivers charge. Repeated charge transfer behaves like current.

$$Q = C_s(V_1 - V_2), \quad I = f_s Q = f_s C_s(V_1 - V_2)$$

Since resistor current is

$$I = \frac{V_1 - V_2}{R_{eq}}$$

comparing gives

$$R_{eq} = \frac{1}{f_s C_s}$$

6.2 What the examiner wants

1. Why SC is useful in IC: resistors occupy area; capacitors and MOS switches are easier.
2. Equivalent resistance formula.
3. Correct replacement diagram.
4. For filter realization, replace every resistor by its own switched capacitor.
5. Mention non-overlapping clocks ϕ_1 and ϕ_2 .

6.3 SC question attack

If the question gives R and f_s , immediately calculate:

$$C_s = \frac{1}{R f_s}$$

Example structure:

$$R = 10 \text{ k}\Omega, \quad f_s = 100 \text{ kHz}$$

$$C_s = \frac{1}{(10 \times 10^3)(100 \times 10^3)} = 10^{-9} \text{ F} = \boxed{1 \text{ nF}}$$

Diagram instruction: Do not draw the switched capacitor block floating. Draw Node 1 $\rightarrow$ switch $\phi_1 \rightarrow$ centre node $\rightarrow$ switch $\phi_2 \rightarrow$ Node 2. Draw C_s from centre node to ground.

7. Phase 3: State-Variable and Universal Filters

7.1 One-line concept

A state-variable filter is called a universal filter because the same active network can provide low-pass, high-pass, band-pass and band-stop outputs by taking output from different nodes.

7.2 Standard denominator

Most second-order filters can be recognized by:

$$D(s) = s^2 + \frac{\omega_0}{Q}s + \omega_0^2$$

For a band-pass filter:

$$H_{BP}(s) = \frac{A_B s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

For bandwidth:

$$B = \frac{\omega_0}{Q}, \quad Q = \frac{\omega_0}{B}$$

7.3 State-variable output idea

Output node	Response obtained
Before first integrator	high-pass output
After first integrator	band-pass output
After second integrator	low-pass output
Sum of HP and LP paths	band-stop / notch output

7.4 Exam derivation pattern

Write the three outputs as:

$$H_{HP}(s) = \frac{K_1 s^2}{s^2 + K_2 K_4 s + K_2 K_3}$$

$$H_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}$$

$$H_{LP}(s) = \frac{K_1 K_2 K_3}{s^2 + K_2 K_4 s + K_2 K_3}$$

Then compare with standard form:

$$\omega_0^2 = K_2 K_3, \quad \frac{\omega_0}{Q} = K_2 K_4$$

$$\boxed{\omega_0 = \sqrt{K_2 K_3}}, \quad \boxed{B = K_2 K_4}$$

8. Phase 4: PLL

8.1 One-line concept

A Phase Locked Loop is a feedback system in which the VCO output phase/frequency follows the input phase/frequency.

8.2 Blocks of PLL

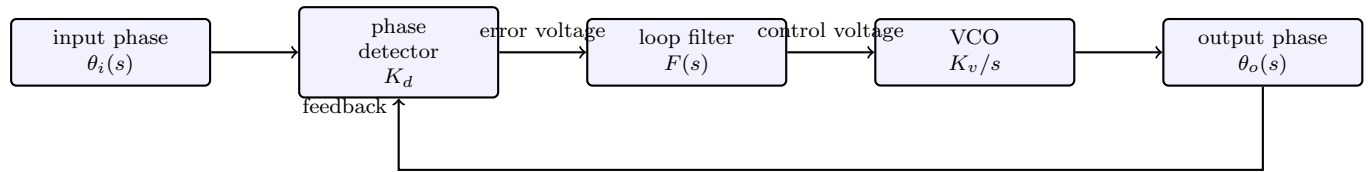


Figure 3: Linear model of PLL with separated labels.

8.3 Derivation you must memorise

Phase error:

$$\phi(s) = \theta_i(s) - \theta_o(s)$$

Phase detector output:

$$V_e(s) = K_d \phi(s)$$

Loop filter output:

$$V_d(s) = F(s) K_d [\theta_i(s) - \theta_o(s)]$$

VCO:

$$\theta_o(s) = \frac{K_v}{s} V_d(s)$$

Substitute:

$$\theta_o(s) = \frac{K_d K_v F(s)}{s} [\theta_i(s) - \theta_o(s)]$$

Let $K = K_d K_v$:

$$\boxed{\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}}$$

8.4 PLL demodulator logic

FM input:

$$\omega_i(t) = \omega_c + k_f m(t)$$

VCO:

$$\omega_o(t) = \omega_c + K_v V_d(t)$$

In lock:

$$\omega_o(t) = \omega_i(t)$$

Therefore:

$$V_d(t) = \frac{k_f}{K_v} m(t)$$

So the loop filter/control voltage is the demodulated output.

9. Phase 5: DAC and ADC

9.1 DAC minimum understanding

DAC converts a digital number into an analog voltage. For an n -bit unipolar DAC:

$$V_o = V_{ref} \frac{D}{2^n}$$

where D is decimal equivalent of the binary input.

LSB size:

$$1 \text{ LSB} = \frac{V_{FS}}{2^n}$$

For R-2R ladder with inverting op-amp:

$$V_o = -V_{ref} \left(\frac{b_{n-1}}{2} + \frac{b_{n-2}}{4} + \cdots + \frac{b_0}{2^n} \right)$$

9.2 DAC error answers

Error	Meaning
Offset error	output is not zero when digital input is zero
Gain error	slope of actual transfer curve differs from ideal slope
Linearity error	maximum deviation of actual output points from ideal straight line after offset and gain correction
Monotonicity	output should never decrease when input code increases
Settling time	time required for output to settle within specified error band after input changes

9.3 ADC minimum understanding

ADC converts analog voltage into digital code. Quantization step:

$$\Delta = \frac{V_{FS}}{2^n}$$

Quantization error:

$$e_q = \pm \frac{1}{2} \text{LSB}$$

9.4 ADC types

ADC		Main idea	Exam point
Counter ADC	type	counter increases DAC output until it reaches input	slow; conversion time depends on input
SAR ADC		tests bits from MSB to LSB by binary search	fixed conversion time nT_{clk}
Dual-slope ADC		integrates input then integrates reference in opposite direction	high noise rejection; used in DVM
Sample and hold		samples analog input and holds constant during conversion	prevents error for changing input

10. Phase 6: Butterworth and Chebyshev Filters

10.1 Butterworth: maximally flat

Butterworth response has no ripple in passband. Normalized magnitude squared:

$$|H(j\omega)|^2 = \frac{1}{1 + \omega^{2n}}$$

Pole location result:

$$\boxed{\sigma^2 + \omega^2 = 1}$$

Hence normalized Butterworth poles lie on a unit circle.

10.2 Chebyshev: equiripple passband

Chebyshev response has passband ripple and sharper transition. Magnitude squared:

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}$$

Poles lie on ellipse:

$$\boxed{\left(\frac{\sigma}{\sinh \alpha}\right)^2 + \left(\frac{\omega}{\cosh \alpha}\right)^2 = 1}$$

10.3 Why Chebyshev -3 dB frequency is greater than 1 rad/s

At $\omega = 1$:

$$|H(j1)|^2 = \frac{1}{1 + \epsilon^2}$$

For usual ripple less than 3 dB, $\epsilon < 1$, so:

$$\frac{1}{1 + \epsilon^2} > \frac{1}{2}$$

Therefore half-power has not yet occurred at $\omega = 1$. Hence:

$$\boxed{\omega_{3dB} > 1 \text{ rad/s}}$$

10.4 Band-pass and band-stop design

For band-pass:

$$H_{BP}(s) = \frac{ABs}{s^2 + Bs + \omega_0^2}$$

where

$$Q = \frac{\omega_0}{B}$$

For band-stop from band-pass with same passband gain A :

$$H_{BS}(s) = A - H_{BP}(s)$$

11. Question Recognition Table

If the question says...	Start with...	Must include
“functions of trigger input”	Trigger synchronizes sweep with input	trigger block, level, slope, stable display
“delay line”	Delay lets time-base start before signal reaches plates	vertical channel diagram
“matched probe”	Probe and CRO input form frequency-dependent divider	$R_p C_p = R_s C_s$ diagram
“Lissajous phase”	X-Y mode, same frequency gives ellipse	$\phi = \sin^{-1}(y_1/y_2)$
“advantages of SC”	resistor replaced by clocked capacitor	$R_{eq} = 1/(f_s C_s)$
“state variable universal”	same circuit gives LP, HP, BP, BS	two integrators and summer
“linear model of PLL”	phase detector, filter, VCO in feedback	$KF(s)/(s + KF(s))$
“PLL demodulator”	locked VCO follows FM input	$V_d(t) = k_f m(t)/K_v$
“R-2R DAC”	binary weighted currents summed by op-amp	3-bit circuit and output table
“linearity error”	deviation from ideal straight line	transfer characteristic graph
“SAR ADC”	binary search using DAC and comparator	flowchart
“dual-slope ADC”	integrate input then reference	timing diagram and formula
“Butterworth poles”	normalized magnitude denominator	unit circle proof
“Chebyshev poles”	ripple polynomial response	ellipse proof

12. Last-48-Hour Practice Protocol

12.1 Three-pass method

Pass 1: Recognition pass

Take all solved Part-II papers. For each question, write only the family name:

CRO / SC / PLL / DAC / ADC / Filter / State-variable

Do this for every question in 30 minutes. The aim is speed of identification.

Pass 2: Skeleton pass

For each question, write only:

1. heading,
2. diagram name,
3. formula,
4. final conclusion.

Do not write full paragraphs.

Pass 3: Full answer pass

Choose the most probable long questions:

- PLL linear model and demodulator,
- R-2R DAC and DAC errors,
- SAR or dual-slope ADC,
- state-variable filter,
- Butterworth/Chebyshev pole proof,
- band-pass/band-stop filter design,
- CRO matched probe / Lissajous / time-base.

Write full answers with diagrams.

12.2 How to self-check

After writing an answer, check these five points:

Check 1: Did I write the direct definition?

Check 2: Is my diagram labelled clearly?

Check 3: Did I define each symbol in the formula?

Check 4: Did I derive or substitute step-by-step?

Check 5: Did I box the final answer?

13. One-Page Formula Bank

CRO

$$V_x(t) = kt, \quad \phi = \sin^{-1} \left(\frac{y_1}{y_2} \right)$$

$$R_p C_p = R_s C_s, \quad C_p = \frac{R_s C_s}{R_p}$$

$$\frac{f_y}{f_x} = \frac{N_h}{N_v}$$

Switched capacitor

$$R_{eq} = \frac{1}{f_s C_s}, \quad C_s = \frac{1}{R_{eq} f_s}$$

PLL

$$\phi(s) = \theta_i(s) - \theta_o(s)$$

$$\frac{\theta_o(s)}{\theta_i(s)} = \frac{K_d K_v F(s)}{s + K_d K_v F(s)}$$

$$V_{demod}(t) = \frac{k_f}{K_v} m(t)$$

DAC/ADC

$$V_o = V_{ref} \frac{D}{2^n}, \quad 1 \text{ LSB} = \frac{V_{FS}}{2^n}$$

$$\Delta = \frac{V_{FS}}{2^n}, \quad e_q = \pm \frac{1}{2} \text{ LSB}$$

Filters

$$Q = \frac{\omega_0}{B}, \quad H_{BP}(s) = \frac{ABs}{s^2 + Bs + \omega_0^2}$$

$$H_{BS}(s) = A - H_{BP}(s)$$

$$|H_B(j\omega)|^2 = \frac{1}{1 + \omega^{2n}}$$

$$|H_C(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}$$

14. Final Exam Writing Commands

Do not write a beautiful but directionless paragraph. Every answer must move in blocks: definition, diagram, equation, derivation, final box.

If stuck in exam: draw the standard diagram first, then write the governing formula. Even if full derivation is incomplete, this recovers marks and also reminds your brain of the answer.

Before entering exam hall

Revise only these pages:

1. Diagram Drawing Command Sheet.
2. Question Recognition Table.
3. One-Page Formula Bank.
4. Phase headings: CRO, SC, PLL, DAC/ADC, Filters.

During exam

Command 1: Spend first 5 minutes marking question families.

Command 2: Attempt the family you know best first.

Command 3: For every long answer, draw first.

Command 4: Keep diagrams inside margins; if needed, turn diagram horizontal.

Command 5: Never leave a numerical without unit.

Command 6: Box the final answer.

Final mindset: You are not memorising pages. You are learning answer-machines.

Recognise the machine, draw it cleanly, write its formula, derive calmly, and finish with a boxed result.